“Improving Web Automation with DOM Delta Processing and Privacy-Aware LLM Agents”

Major Project – 2

Computer Science & Engineering (Data Science)

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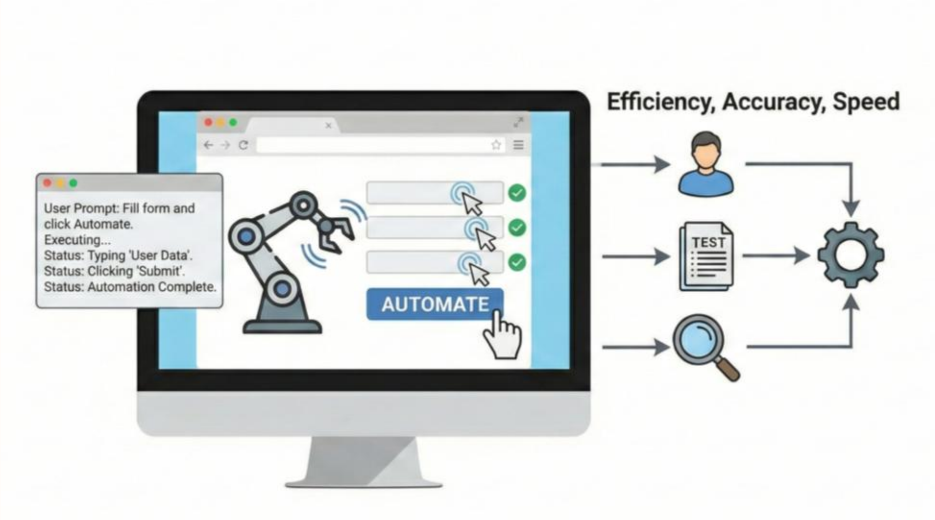
Contents

Introduction

- GUI automation is a way to automate human interaction such as click, type, scroll, input etc. with user given instruction.
- GUI automation accelerates tasks in **testing**, **verification**, **bug replay**, and **text-input generation**, reducing human effort and improving reliability.
- Natural language interfaces allow users to describe tasks (“Order grocery”, “Change Settings”), enabling automatic execution by intelligent agents.
- Automation is essential because manual GUI operations are slow, repetitive, error-prone, and difficult to scale.



Problem Statement

- Most LLM-based web-agents are based on entire DOM and at every steps uses 10^4–10^5 tokens, which is computationally expenses, even as most time important interactive elements are only in updated DOM from last step.
- Most element grounding methods are either used cloud-based solution or large models which are not capable to perform without GPUs.
- Lack of Privacy and HITL safeguards in web-automations.
- A localised, token-efficient, privacy-aware web agent that runs on commodity hardware is still missing.

Literature Review

Year	Title	Key Contribution	Limitations
May, 2025 [1]	Large Language Model-Brained GUI Agents: A Survey	Systematic , detailed survey of LLM-brained GUI agents: historical evolution, core components, datasets, evaluation metrics, applications. acts as a "cookbook" for building GUI agents.	Identifies open research gaps : robust grounding, benchmarks, long-horizon planning.
Jun, 2025 [2]	A Survey on (M)LLM-Based GUI Agents	Breaks down modern GUI agents into four components (perception, exploration, planning, interaction) and analyzes technical/evaluation limitations.	Notes methodological limitations in current benchmarks and evaluation practices also highlight for standardization and improved element localization.
Aug, 2025 [3]	A Survey of WebAgents: Towards Next-Generation AI Agents for Web Automation with Large Foundation Models	Reviews WebAgent architectures, training strategies, and trustworthiness. frames web automation as a productivity/UX problem enabled by LFMs.	Highlights trustworthiness and robustness issues, and the need for better training/regulatory practices for WebAgents.

Literature Review

Year	Title	Key Contribution	Limitations
Feb, 2025 [4]	GUI Agents with Foundation Models: A Comprehensive Survey	Consolidates recent work using (M)LLMs for GUI tasks. offers taxonomy, datasets review, and unified framework.	Discusses practical gaps like unavailable checkpoints, inconsistent experimental settings, and reproducibility challenges.
Mar, 2024 [5]	GPT-4V(ision) is a Generalist Web Agent, if Grounded	Shows GPT-4V (multimodal LMM) can act as a generalist web agent when grounded with both screenshot + HTML; introduces SeeAct, evaluates on MIND2WEB (51.1% success with manual grounding).	grounding strategies are imperfect and there's a significant gap to oracle grounding. SeeAct's best gains rely on expensive LMMs and manual grounding infrastructure.
Nov, 2024 [6]	AdaptAgent: Adapting Multimodal Web Agents with Few-Shot Learning from Human Demonstrations	Proposes few-shot adaptation using human demonstrations to improve web agents' generalization on unseen sites. shows 3.36%–7.21% absolute gains (relative 21%–66%) on benchmarks.	Improvement depends on demonstration selection and model type. few-shot gains vary and do not fully close the generalization gap for proprietary or very different sites.

Literature Review

Year	Title	Key Contribution	Limitations
May, 2025 [7]	LiteWebAgent: The Open-Source Suite for VLM-Based Web-Agent Applications	Provides an open, modular framework (planning, memory, tree search) plus production demos (serverless backend or Chrome extension) to help researchers build and deploy VLM web agents.	actual performance depends on chosen models and grounding modules.
Aug, 2024 [8]	OpenWebAgent: An Open Toolkit to Enable Web Agents on Large Language Models	Delivers a modular toolkit with an HTML parser, action generation module, and UI to accelerate building of LLM/LMM web agents; open-source code & plugin.	Toolkit-oriented – does not itself claim new agent performance. Practical utility depends on integration choices and model quality.
Jul, 2025 [9]	Browsing Like Human: A Multimodal Web Agent with Experiential Fast-and-Slow Thinking (WebExperT)	Proposes a human-like fast/slow planning paradigm and experiential learning (reflecting on failures) to improve decomposition of complex instructions – shows strong results on MIND2WEB.	Added complexity (planning + reflection) may increase compute and implementation complexity.

Literature Review

Year	Title	Key Contribution	Limitations
May, 2025 [11]	Prune4Web: DOM Tree Pruning Programming for Web Agent	Three-stage Planner→Filter→Grounder; programmatic JS-extractor + chunked LLM judge; 88.28% EA Mind2Web with fine-tuned Qwen2.5VL-3B-Instruct grounder.	Filter is prompt-engineered & LLM-bound. grounder is a 3B VLM and no privacy/HITL layer
2025 [12]	Beyond Pixels: Exploring DOM Downsampling for LLM-Based Web Agents	DOM-diff snapshotting; ~10^3-token compression via post-order merging + TextRank	Operates on isolated snapshots; no cross-step reuse
2024 [13]	Agent-E: From Autonomous Web Navigation to Foundational Design Principles in Agentic Systems	Accessibility-tree pruning; MutationObserver for cross-step changes	Converts changes into prompt text — no token reduction

Literature Review

Year	Title	Key Contribution	Limitations
2024 [14]	AutoWebGLM: A Large Language Model-based Web Navigating	HTML pruning by trimming non-actionable nodes, depth-limit	Per-step snapshot only
2023 [15]	A Real-World Webagent With Planning, Long Context Understanding, And Program Synthesis	HTML-specific encoder for long DOM sequences	Each step independent & no delta reuse
2020 [16]	MINILM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers	22M-param distilled encoder, ms-level CPU inference	Generic encoder, not DOM-aware

Literature Review

Year	Title	Key Contribution	Limitations
2023 [17]	QLORA: Efficient Finetuning of Quantized LLMs	4-bit nf4 + LoRA — billion-scale fine-tuning on a single consumer GPU	—

Research Gaps

Unreliable Element Grounding

- Current agents struggle with fine-grained visual-to-HTML grounding, causing major failures on real websites [4] [5] [9].

Stateless full-DOM filtering

- Every existing pruning system (Prune4Web, AutoWebGLM, Agent-E, HTML-T5) re-processes the full DOM each step, even when the page changed marginally. No structural-UID + delta layer is wired into a real filter [11] [13] [14] [15].

Sub-1B local grounders are under-studied

- Most web agents dependent on GPT-4V or Qwen2.5VL-3B. Also open-weight small models such as 0.5B grounders exist with not having strong number [11].

Privacy and HITL Limitations

- Mosts Web-automation tools, do not look for sensitive pages like CAPTCHA, Payment-Gateway, login, or explicit privacy policies at each subtask level [1] [3].

Objectives

- Build a web automation agent using multimodal which can perform better on some parameter to improve the overall automation process.
- Use HTML summarizer which can parse only query related values further we can use that as input to large models.
- Add DOM Delta layer for better inference with filter stage with MiniLM based scoring.
- Replace closed Qwen2.5-3B grounder with QLORA fine-tuned 0.5B or 0.6B model.
- Add Policy HUB and HITL layers with customizable policies and keywords.

Existing Solution

Paper-Title: Prune4Web: DOM Tree Pruning Programming for Web Agent

- Three-stage pipeline: Planner (decides sub-task) → Programmatic Element Filter (LLM emits a Python keyword:weight script that a deterministic scorer applies) → Action Grounder (fine-tuned Qwen2.5VL-3B-Instruct).
- Reports 88.28% Element Accuracy on Mind2Web test_task with the fine-tuned 3B VLM grounder also filter is having 25 to 50 times DOM-token reduction at the filter stage.
- Chosen as base over SeeAct because SeeAct depends on GPT-4V (paid) and could not be reproduced locally while Prune4Web has an local inference capability and a stronger published number.

Source: [5] [11]

Dataset: MultiModal-Mind2Web

What is Multimodal-Mind2Web:

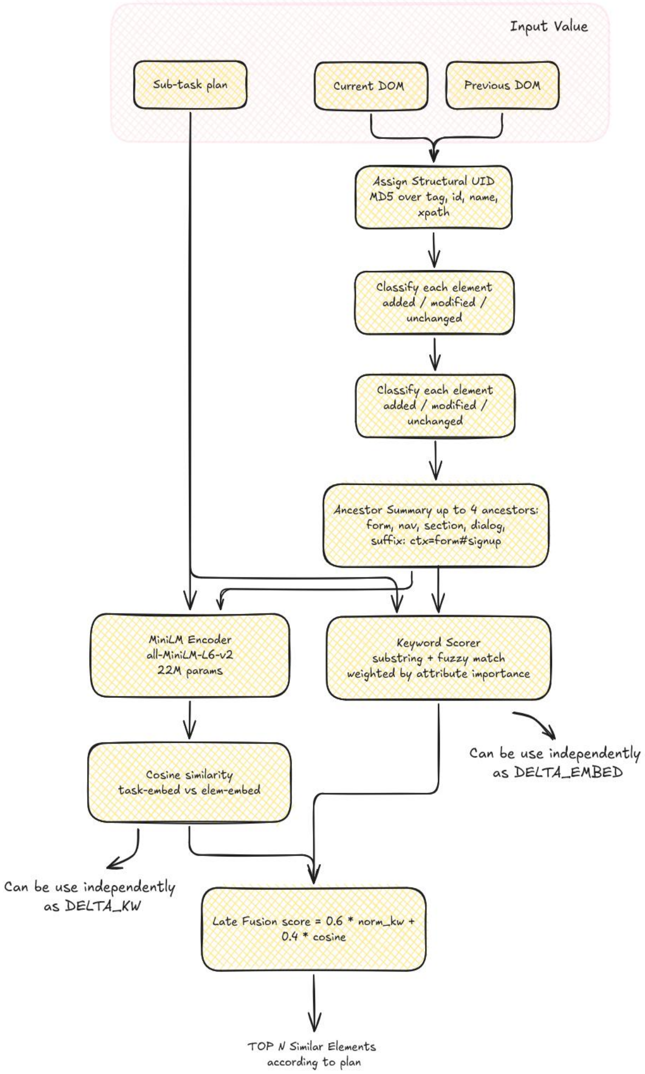
- It is the multimodal version of Mind2Web, Where each webpage state has both screenshot image + HTML DOM [10].
- Covers 137 websites, spanning 31 top-level domains, with a broad variety of user tasks + interaction patterns [10].
- Consists of over 2,000+ distinct tasks (in the original Mind2Web) with crowdsourced action sequences [10].

Split / Set	No. of actions / steps	What it means
Train	7775 actions from 1009 tasks	training pool for learning / experimentation
Test_task	1339 actions from 177 tasks	tests generalization across tasks within known websites
Test_website	1019 actions from 142 tasks	tests cross-website generalization
Test_domain	4060 actions from 694 tasks	tests cross-domain robustness

Source: [10]

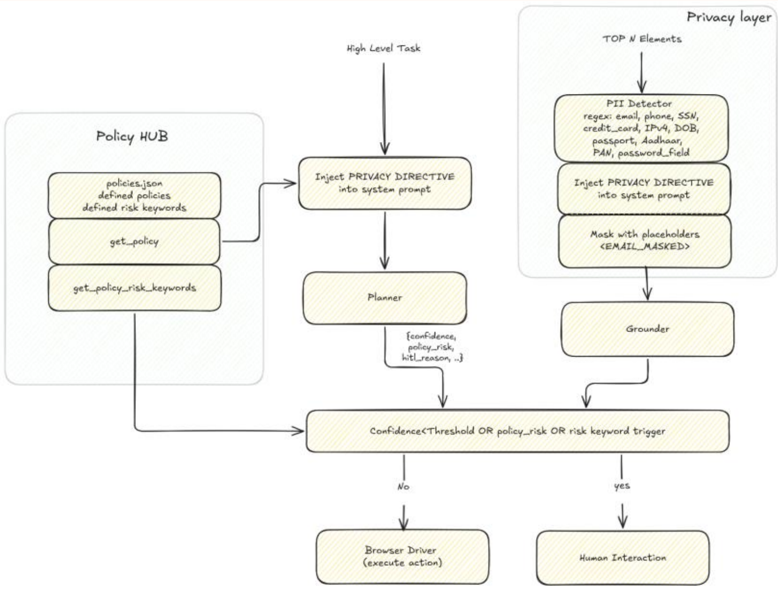
Novelty 1: DOM Delta Processing

- **Structural UID:** tag, id, name, xpath values are stable across re-renders, even when visible text mutates. Same property as Mind2Web's backend_node_id but computed offline from parsed HTML, where no live browser needed.
- **Ancestor summary:** Keep up to 4 semantically meaningful ancestors (form, nav, section, dialog) appended which preserve ancestor summary.
- **Hybrid score:** calculate hybrid score with delta keyword and with delta sentence embeddings from all-MiniLM-L6-v2.
- **Result:** As a result 16–18% Qwen-token reduction at matched Recall@20 across all three test splits.



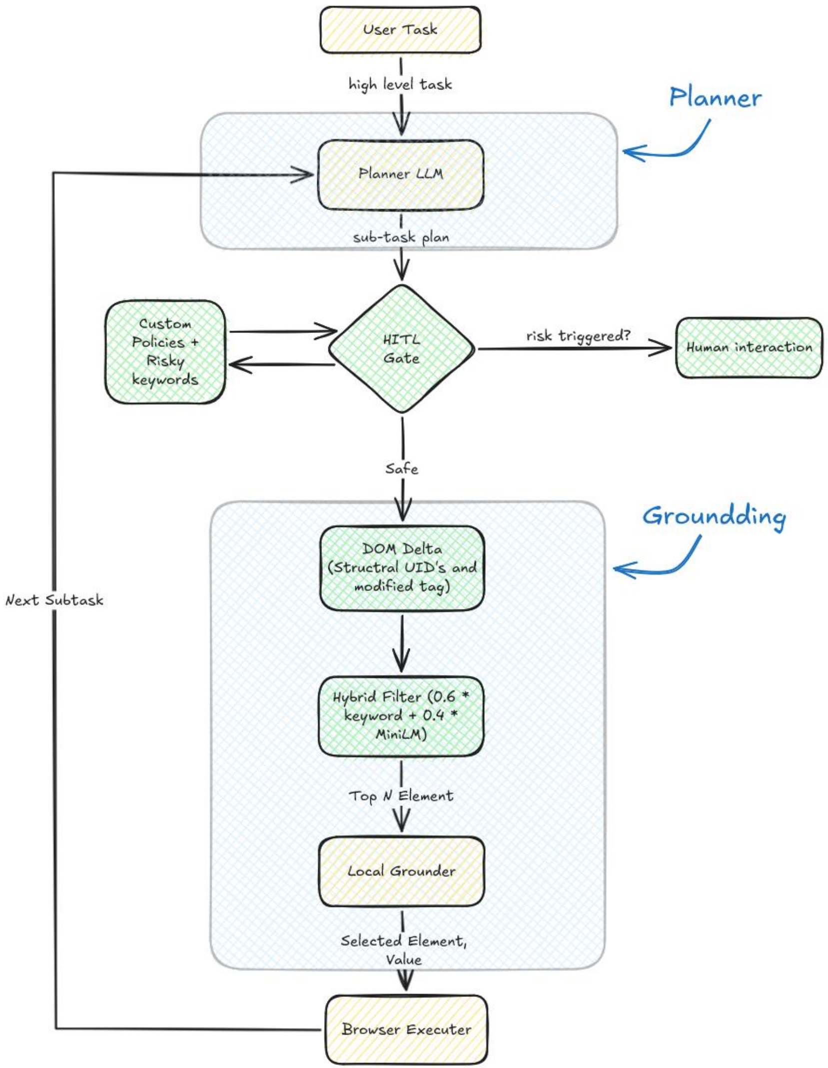
Novelty 2: Policy Hub + Risk-Based HITL

- **PolicyHub:** Contain Policies and risky keywords list, which is customizable and can contain complete policy as well.
- LLM JSON output extended with policy_risk (bool), risk_confidence, hitl_reason.
- **HITL gate** fires on any one of three OR-combined triggers: (a) LLM self-flags policy_risk=true, (b) a risk keyword appears in the action text or hitl_reason, (c) risk_confidence falls below a configured threshold.
- The gate sits before the grounder, so risky steps further inference can be avoided by the expensive grounder call.
- **Privacy-Aware LLM** wrapper (between Filter and Grounder) runs a regex PII detector such as email, phone, Aadhaar, PAN, password and replaces matches with masking [EMAIL_MASKED] style placeholders. A short privacy directive is injected into the grounder system prompt instructing the model to treat masked tokens as opaque.



Architecture

- **Action Generation** (LMM): Given the current screenshot + task + history to model which return a textual action and risk scores for action.
- **Action Grounding**: Map textual description to actual DOM element + operation (click / type / select) by small model and with dom delta methodology.
- **Browser Execution**: Use automation (via Playwright / browser engine) to execute the action. Webpage updates and we get new screenshot + DOM.



Improvement

- **Prune4Web:**

$a_t = \pi(s_t, T, \{a_1, a_2, \ldots, a_{t-1}\})$

$s_{t+1} = S(a_t) = \{h_{t+1}, i_{t+1}\}$

Where:
a_t = action at time t
s = site specific input (html + image)
π = Multimodel LLM
T = user defined task

- **Our's:**

$A_t = \pi(s_t, T, \{a_1, a_2, \ldots, a_{t-1}\}, p_t)$

$s_{t+1} = S(a_t) = \{h_{t+1}, i_{t+1}\}$

Where:
a_t = action at time t
s = site specific input (html + image)
π = Multimodel LLM
T = user defined task
p_t = Policy rules
$A_t \in (a_t, c, r, hr)$
c = confidence $c \in [0,100]$,
$r \in \{0,1\}$,
hr = hitl_reason

Improvement - Algorithm

```
At = π(st, T, {a1, a2, ..., at−1}, pt)
at, c, r, hr = At

matched = compare action plan and policy keywords

if (r>0 or matched or c<confidence){
 r = 1 #policy_risk detected
}

if (r == 1) {
 HITL value triggered, as wait for human interaction for step
} else {
 Executer further steps from grounding
}
```

Implementation Notes

- Our architecture inspired from Prune4Web with some deviations.
- Grounder: QLoRA-fine-tuned Qwen2.5-0.5B or Qwen3-0.6B loaded in 4-bit nf4; <1 GB VRAM at inference on RTX 4050 (6 GB).
- Fine-tuned on 6,626 Mind2Web examples using the identical grounder prompt as inference — no train/inference drift.
- DOM Delta + Privacy wrapper run entirely on-device.

Experiments: Filter-Stage Recall@20

Split	Samples	FULL Recall@20	DELTA_HYBRID Recall@20	Δ	Token red.
test_task	1,257	90.30%	90.07%	-0.23 pp	17.8%
test_website	975	89.95%	89.85%	-0.10 pp	16.4%
test_domain	3,838	86.78%	87.98%	+1.20 pp	17.5%

DELTA_HYBRID matches FULL Recall@20 within 0.23 pp on two splits and improves on the largest split, while reducing grounder input tokens by 16–18%.

Experiments: Grounder Element Accuracy

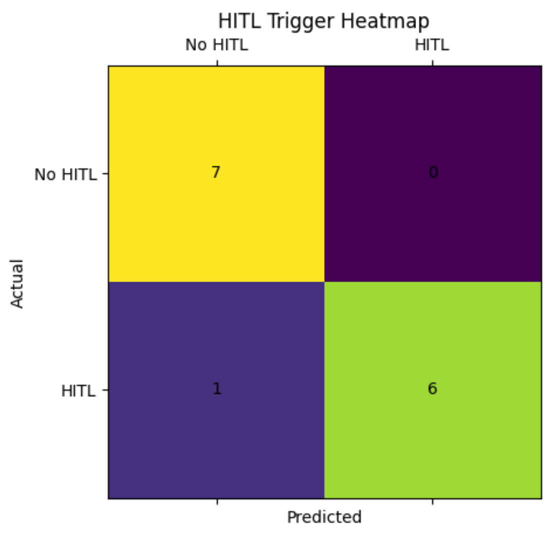
Grounder	Params	test_task EA	Notes
Qwen2.5VL-3B-Instruct (Prune4Web)	3.0B	88.28%	Vision-language, paper baseline
Qwen2.5-0.5B + QLoRA (ours)	0.5B	85.00%	Text-only, 6× fewer params
Qwen3-0.6B + QLoRA (ours)	0.6B	88.00%	Text-only, 5× fewer params, matches paper

200-sample test_domain slice with DOM Delta enabled: Qwen3 reaches 88.00% EA, matching the 3B vision-language baseline within rounding margin while running without a vision encoder.

Implementation & Initial Result

*Initial results are computed on 13 live websites (7 required HITL, 6 not required HITL)

Metric	Value
HITL Recall	85.71%
HITL Precision	100%
HITL Accuracy	92.86%
False Positive Rate	0%
False Negative Rate	14.29%



- The system is able in triggering HITL.
- It avoids unnecessary human interruptions (0 false positives),
- but misses some high-risk cases (false negatives),
- which we aim to improve via expanded risk keywords and site-level overrides.

Future Work

- Future work should focus on gathering data for evaluating PolicyHub and privacylayer on large scale.
- Even though Recall@20 of FULL and DELTA HYBRID is 1 pp we observed stable 5 to 6 , pp Element Accuracy gap between same QLoRA grounder and 3B model, even though their differs by only on the same 200 test domain samples.
- Experiment should try to reorder top-20 block before grounding and to re-measure EA with both grounders. If the gap closes, candidate-order normalisation should be folded back into the DOM Delta layer. if it does not, the grounder is sensitive to the score-induced ordering itself, which would motivate a small preference-tuning around on ordered candidate blocks.

Conclusion

- DOM Delta Processing replaces the stateless full-DOM filter with a structural-UID-tracked delta view. hybrid keyword + MiniLM scoring gives 16–18% token reduction at matched or improved Recall@20 across 6,070 Mind2Web test samples.
- A QLoRA-fine-tuned Qwen3-0.6B local grounder reaches 88.00% EA, matching the Prune4Web 3B Qwen2.5VL-Instruct baseline (88.28%) within rounding margin while running at 5× fewer parameters, no vision encoder, <1 GB VRAM on a 6 GB RTX 4050.
- A PolicyHub + risk-based HITL gate with a regex PII detector and a privacy directive injected into the grounder prompt brings privacy-awareness into the action loop, not bolted on after.
- Net result: the Prune4Web pipeline becomes cheaper at the filter stage, safer at the grounder stage, and deployable on consumer hardware without sacrificing element accuracy.

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Do have any questions?

Thanks!